King Saud University
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Group Term Project: Lexical Analyzer

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Pledge:

We pledge that this project write-up and coding is our own work.

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1 Data Structures Used

To implement the token recognizer pipeline from scratch, our Python implementation relies on a custom directed graph architecture.

- **State Class:** Represents a single node in our automata. It tracks:
  - `id`: A unique integer for identification.
  - `transitions`: A dictionary (Hash Map) mapping a character (e.g., 'a') to a set of destination `State` objects.
  - `e_transitions`: A set of `State` objects reachable via ϵ-transitions (without consuming input).
  - `is_accepting`: A boolean flag. If true, the state also stores `token_type` (a string like "KW_IF") and `priority` (an integer denoting its order in the specification list to resolve ties).
- **NFA Class:** A lightweight wrapper that holds pointers to a `start_state` and an `accept_state` to facilitate modular Thompson's Construction.

2 Algorithms in Pseudocode

2.1 Regex Handling & NFA Construction (Thompson's)

The parser explicitly handles implicit concatenation, grouping `()`, union `|`, Kleene star `*`, plus `+`, and optional `?`.

```
ALGORITHM RegexToNFA(regex_string):
    tokens = InsertImplicitConcatenation(regex_string)
    postfix = ConvertToPostfix(tokens) // Using Shunting-Yard
    stack = EmptyStack()

    FOR EACH token IN postfix:
        IF token is Operator ('*', '+', '?', '|', '.'):
            Pop necessary NFA(s) from stack
            Apply Thompson's Construction rule (e.g., Union,
  Concat, Star)
            Push resulting NFA to stack
        ELSE:
            Push NFA for single character or range to stack

    RETURN stack.pop()
```

2.2 Subset Construction (NFA to DFA)

This algorithm flattens the non-deterministic graph into a deterministic one, processing the "Super NFA" created by linking all token NFAs to a master start state with ϵ-transitions.

```
ALGORITHM BuildDFA(super_nfa_start):
    initial_closure = e_closure({super_nfa_start})
    unmarked_states = Queue([initial_closure])
    dfa_transitions = Map()
    dfa_accepting = Map()

    WHILE unmarked_states is not empty:
        current_nfa_set = unmarked_states.dequeue()
        current_dfa_id = Hash(current_nfa_set)

        // Priority Tie-Breaking Rule
        best_priority = INFINITY
        winning_token = NULL

        FOR state IN current_nfa_set:
            IF state.is_accepting AND state.priority <
   best_priority:
                best_priority = state.priority
                winning_token = state.token_type

        IF winning_token is not NULL:
            dfa_accepting[current_dfa_id] = winning_token

        FOR EACH symbol in vocabulary:
            next_set = e_closure(move(current_nfa_set, symbol))
            IF next_set is not empty:
                next_dfa_id = Hash(next_set)
                IF next_dfa_id is entirely new:
                    unmarked_states.enqueue(next_set)
                dfa_transitions[current_dfa_id][symbol] =
   next_dfa_id

    RETURN dfa_start, dfa_transitions, dfa_accepting
```

2.3 Tokenization Strategy (Maximal Munch)

This algorithm ensures the scanner always consumes the longest possible matching prefix.

```
ALGORITHM SimulateLexer(source_code, dfa):
    current_pos = 0
    WHILE current_pos < length(source_code):
        Skip whitespace characters
        current_state = dfa.start
        last_accept_state = NULL
        last_accept_pos = -1
        lookahead = current_pos

        WHILE lookahead < length(source_code):
            char = source_code[lookahead]
            IF valid transition exists from current_state on char
   :
```

```
            current_state = transition(current_state, char)
            IF current_state in dfa_accepting:
                last_accept_state = current_state
                last_accept_pos = lookahead
            lookahead++
        ELSE:
            BREAK // DFA Crashed

    IF last_accept_state is not NULL:
        token = dfa_accepting[last_accept_state]
        lexeme = source_code[current_pos to last_accept_pos]
        PRINT lexeme, token, position
        current_pos = last_accept_pos + 1
    ELSE:
        PRINT "Lexing Error" at current_pos
        HALT
```

3 Cost Analysis

3.1 Time Complexity

- **NFA Construction:** $\mathcal{O}(|R|)$ where $|R|$ is the length of all regular expressions combined. The Shunting-Yard and Thompson's algorithms process each character a constant number of times.
- **Subset Construction:** $\mathcal{O}(2^{|Q|})$, where $|Q|$ is the number of NFA states. In the worst case, a DFA state could represent any possible subset of NFA states.
- **Simulation:** $\mathcal{O}(N)$, where N is the length of the input source code. Dictionary lookups for transitions execute in constant time $\mathcal{O}(1)$.

3.2 Space/Memory Complexity

- **NFA:** $\mathcal{O}(|R|)$ space, as Thompson's rules add a small, fixed maximum number of states per operator.
- **DFA:** $\mathcal{O}(2^{|Q|} \times |\Sigma|)$ worst-case space to store the flattened transition table across the input alphabet.
- **Simulation:** $\mathcal{O}(1)$ auxiliary space, as it relies on simple positional pointers and single-state tracking regardless of file size.

4 Sample Runs

4.1 Sample Run 1: Full Pipeline (Target Test String)

This run demonstrates the complete pipeline parsing the complex test string provided in the project assignment, resolving conflicts using Priority and Maximal Munch.

Step 1: Input Specifications (Option 2) & Compilation (Option 3)

```
("KW_IF", "if"),
("KW_THEN", "then"),
("KW_ELSE", "else"),
("ID", "[A-Za-z][A-Za-z0-9_]*"),
("NUM", "[0-9]+(\.[0-9]+)?"),
("EQ", "=="),
("ASSIGN", "="),
("OP_PLUS", "\+"),
("OP_MINUS", "\-"),
("OP_MULT", "\*"),
("OP_DIV", "/"),
("OP_NEQ", "!="),
("OP_LEQ", "<="),
("OP_GEQ", ">="),
("OP_LT", "<"),
("OP_GT", ">"),
("LBRACE", "\{"),
("RBRACE", "\}"),
("LPAREN", "\("),
("RPAREN", "\)"),
("SEMI", ";"),
("COMMA", ",")

Select an option: 3
DFA Compilation Successful.
```

Step 2: Input Source Code & Token Stream Output (Option 4)
Input: `if x > y then {x=y; z=x-1} else x=y+5`

```
Lexeme          | Token           | Position
-------------------------------------------------
'if'            | KW_IF           | Line 1, col 1
'x'             | ID              | Line 1, col 4
'>'             | OP_GT           | Line 1, col 6
'y'             | ID              | Line 1, col 8
'then'          | KW_THEN         | Line 1, col 10
'{'             | LBRACE          | Line 1, col 15
'x'             | ID              | Line 1, col 16
'='             | ASSIGN          | Line 1, col 17
'y'             | ID              | Line 1, col 18
';'             | SEMI            | Line 1, col 19
'z'             | ID              | Line 1, col 21
'='             | ASSIGN          | Line 1, col 22
'x'             | ID              | Line 1, col 23
'-'             | OP_MINUS        | Line 1, col 24
'1'             | NUM             | Line 1, col 25
'}'             | RBRACE          | Line 1, col 26
'else'          | KW_ELSE         | Line 1, col 28
'x'             | ID              | Line 1, col 33
'='             | ASSIGN          | Line 1, col 34
'y'             | ID              | Line 1, col 35
```

```
'+'             | OP_PLUS         | Line 1, col 36
'5'             | NUM             | Line 1, col 37
```

(Note: The assignment document lists the final 'x' token starting at column 30, which mathematically overlaps with the 4-letter word 'else' starting at 28. Because our algorithm accurately tracks whitespace, it dynamically calculates the true mathematical offset at column 33).

4.2 Sample Run 2: Union Rule and Error Handling

This independent run uses a smaller, distinct specification list to demonstrate the parser's ability to handle the Union operator (|) and its ability to properly detect and report Lexing Errors when an unrecognized character is encountered.

Step 1: Input Specifications (Option 2) & Compilation (Option 3)

```
("TYPE", "int|float"),
("VAR", "[A-Za-z]+"),
("NUM", "[0-9]+"),
("END", ";")

Select an option: 3
DFA Compilation Successful.
```

Step 2: Input Source Code & Token Stream Output (Option 4)
Input: `int count ; float 5.5`

```
Lexeme          | Token           | Position
-------------------------------------------------
'int'           | TYPE            | Line 1, col 1
'count'         | VAR             | Line 1, col 5
';'             | END             | Line 1, col 11
'float'         | TYPE            | Line 1, col 13
'5'             | NUM             | Line 1, col 19
[!] Lexing Error: Unrecognized char '.' at Line 1, Col 20
```

5 Conclusion

Supported Features: The custom lexical analyzer successfully complies with all pipeline requirements without using built-in regex libraries. It features an internal regex parser supporting grouping, character ranges, multiple closures (`*`, `+`, `?`), union, and escape sequences. The simulation engine strictly enforces Maximal Munch and resolves conflicts (e.g., an `ID` vs. `KW_IF` collision) using a pre-compiled priority tie-breaker system based on input order.

Limitations: The primary limitation is the potential exponential growth associated with Subset Construction. Highly ambiguous regular expressions with massive nondeterminism could lead to state explosion, resulting in slow DFA compilation times and high memory usage. Furthermore, error handling halts at the first invalid character rather than employing a panic-mode recovery mechanism.

Possible Future Improvements: In the future, the lexer could be optimized by applying Hopcroft's Algorithm to minimize the generated DFA, reducing the memory footprint. Additionally, implementing error recovery strategies (such as discarding unidentifiable characters until a recognized delimiter is reached) would allow the scanner to report multiple lexical errors in a single pass.

Acknowledgements: We consulted online documentation to clarify our understanding of some concepts mentioned in this report (e.g., the Shunting-Yard algorithm, Thompson's Construction, Python's `frozenset` data structure, and Hopcroft's algorithm as a proposed future optimization).

Appendix A: Source Code Listing

```
import sys

# ==========================================
# 1. DATA STRUCTURES (STATE & NFA)
# ==========================================

class State:
    _id_counter = 0

    def __init__(self):
        self.id = State._id_counter
        State._id_counter += 1
        self.transitions = {}
        self.e_transitions = set()
        self.is_accepting = False
        self.token_type = None
        self.priority = float('inf')

    def add_transition(self, char, state):
        if char not in self.transitions:
            self.transitions[char] = set()
        self.transitions[char].add(state)


class NFA:
    def __init__(self, start_state, accept_state):
        self.start = start_state
        self.accept = accept_state


# ==========================================
# 2. THOMPSON'S CONSTRUCTION PRIMITIVES
# ==========================================

def build_char_nfa(char):
    start, accept = State(), State()
    start.add_transition(char, accept)
    return NFA(start, accept)


def build_range_nfa(chars):
    start, accept = State(), State()
    for c in chars:
        start.add_transition(c, accept)
    return NFA(start, accept)


def build_concat_nfa(nfa1, nfa2):
    nfa1.accept.e_transitions.add(nfa2.start)
    nfa1.accept.is_accepting = False
    return NFA(nfa1.start, nfa2.accept)


def build_union_nfa(nfa1, nfa2):
    start, accept = State(), State()
    start.e_transitions.update([nfa1.start, nfa2.start])
```

```
    nfa1.accept.e_transitions.add(accept)
    nfa2.accept.e_transitions.add(accept)
    nfa1.accept.is_accepting = False
    nfa2.accept.is_accepting = False
    return NFA(start, accept)


def build_star_nfa(nfa):
    start, accept = State(), State()
    start.e_transitions.update([nfa.start, accept])
    nfa.accept.e_transitions.add(accept)
    accept.e_transitions.add(start)
    nfa.accept.is_accepting = False
    return NFA(start, accept)


def build_plus_nfa(nfa):
    start, accept = State(), State()
    start.e_transitions.add(nfa.start)
    nfa.accept.e_transitions.add(accept)
    accept.e_transitions.add(start)
    nfa.accept.is_accepting = False
    return NFA(start, accept)


def build_optional_nfa(nfa):
    start, accept = State(), State()
    start.e_transitions.update([nfa.start, accept])
    nfa.accept.e_transitions.add(accept)
    nfa.accept.is_accepting = False
    return NFA(start, accept)


# ==========================================
# 3. REGEX PARSER (Infix -> Postfix -> NFA)
# ==========================================

def expand_range(range_str):
    chars = set()
    if range_str == "[A-Za-z]":
        chars.update(chr(i) for i in range(65, 91))
        chars.update(chr(i) for i in range(97, 123))
    elif range_str == "[0-9]":
        chars.update(str(i) for i in range(10))
    elif range_str == "[A-Za-z0-9_]":
        chars.update(chr(i) for i in range(65, 91))
        chars.update(chr(i) for i in range(97, 123))
        chars.update(str(i) for i in range(10))
        chars.add('_')
    else:
        chars.update(list(range_str.strip('[]')))
    return list(chars)


def insert_implicit_concat(regex):
    result = []
    i = 0
    while i < len(regex):
```

```
        char = regex[i]
        if char == '[':
            end = regex.find(']', i)
            result.append(regex[i:end + 1])
            i = end + 1
        elif char == '\\':
            result.append(regex[i:i + 2])
            i += 2
        else:
            result.append(char)
            i += 1

    final_res = []
    for i in range(len(result)):
        final_res.append(result[i])
        if i + 1 < len(result):
            c1, c2 = result[i], result[i + 1]
            if c1 not in ['(', '|'] and c2 not in [')', '|', '*', '+',
    '?']:
                final_res.append('.')
    return final_res


def to_postfix(tokens):
    precedence = {'*': 3, '+': 3, '?': 3, '.': 2, '|': 1}
    output, stack = [], []
    for token in tokens:
        if token in ['*', '+', '?', '.', '|']:
            while (stack and stack[-1] != '(' and
                   precedence.get(stack[-1], 0) >= precedence.get(token
    , 0)):
                output.append(stack.pop())
            stack.append(token)
        elif token == '(':
            stack.append(token)
        elif token == ')':
            while stack and stack[-1] != '(':
                output.append(stack.pop())
            stack.pop()
        else:
            output.append(token)
    while stack:
        output.append(stack.pop())
    return output


def regex_to_nfa(regex_str):
    tokens = insert_implicit_concat(regex_str)
    postfix = to_postfix(tokens)
    stack = []

    for token in postfix:
        if token == '*':
            stack.append(build_star_nfa(stack.pop()))
        elif token == '+':
            stack.append(build_plus_nfa(stack.pop()))
        elif token == '?':
            stack.append(build_optional_nfa(stack.pop()))
```

```
        elif token == '.':
            n2, n1 = stack.pop(), stack.pop()
            stack.append(build_concat_nfa(n1, n2))
        elif token == '|':
            n2, n1 = stack.pop(), stack.pop()
            stack.append(build_union_nfa(n1, n2))
        elif token.startswith('['):
            stack.append(build_range_nfa(expand_range(token)))
        elif token.startswith('\\'):
            stack.append(build_char_nfa(token[1]))
        else:
            stack.append(build_char_nfa(token))

    return stack.pop()


# ==========================================
# 4. SUBSET CONSTRUCTION (NFA -> DFA)
# ==========================================

def e_closure(states):
    stack, closure = list(states), set(states)
    while stack:
        curr = stack.pop()
        for nxt in curr.e_transitions:
            if nxt not in closure:
                closure.add(nxt)
                stack.append(nxt)
    return closure


def move(states, symbol):
    reachable = set()
    for s in states:
        if symbol in s.transitions:
            reachable.update(s.transitions[symbol])
    return reachable


def build_dfa(super_start):
    init_closure = e_closure({super_start})
    dfa_start_id = frozenset(init_closure)

    unmarked = [init_closure]
    dfa_map = {dfa_start_id: init_closure}
    dfa_trans, dfa_accept = {}, {}

    while unmarked:
        curr_nfa_states = unmarked.pop(0)
        curr_dfa_id = frozenset(curr_nfa_states)
        dfa_trans[curr_dfa_id] = {}

        best_pri, winning_tok = float('inf'), None
        for s in curr_nfa_states:
            if s.is_accepting and s.priority < best_pri:
                best_pri = s.priority
                winning_tok = s.token_type
        if winning_tok:
```

```
            dfa_accept[curr_dfa_id] = winning_tok

        symbols = set()
        for s in curr_nfa_states:
            symbols.update(s.transitions.keys())

        for sym in symbols:
            nxt_nfa_states = e_closure(move(curr_nfa_states, sym))
            if not nxt_nfa_states: continue

            nxt_dfa_id = frozenset(nxt_nfa_states)
            if nxt_dfa_id not in dfa_map:
                dfa_map[nxt_dfa_id] = nxt_nfa_states
                unmarked.append(nxt_nfa_states)
            dfa_trans[curr_dfa_id][sym] = nxt_dfa_id

    return dfa_start_id, dfa_trans, dfa_accept


# ==========================================
# 5. SIMULATION (MAXIMAL MUNCH)
# ==========================================

def simulate(source, dfa_start, dfa_trans, dfa_accept):
    pos, line, col = 0, 1, 1
    print(f"\n{'-' * 50}\n{'Lexeme':<15} | {'Token':<15} | Position\n
    {'-' * 50}")

    while pos < len(source):
        char = source[pos]
        if char in ' \t\n\r':
            if char == '\n':
                line += 1
                col = 1
            else:
                col += 1
            pos += 1
            continue

        curr_dfa = dfa_start
        last_acc_state, last_acc_pos = None, -1
        lookahead = pos

        while lookahead < len(source):
            l_char = source[lookahead]
            if curr_dfa in dfa_trans and l_char in dfa_trans[curr_dfa]:
                curr_dfa = dfa_trans[curr_dfa][l_char]
                if curr_dfa in dfa_accept:
                    last_acc_state = curr_dfa
                    last_acc_pos = lookahead
                lookahead += 1
            else:
                break

        if last_acc_state:
            lexeme = source[pos:last_acc_pos + 1]
            tok = dfa_accept[last_acc_state]
            print(f"'{lexeme}'{'' :<{13 - len(lexeme)}} | {tok:<15} |
```

```
    Line {line}, col {col}")
286             col += len(lexeme)
287             pos = last_acc_pos + 1
288         else:
289             print(f"[!] Lexing Error: Unrecognized char '{source[pos]}'
     at Line {line}, Col {col}")
290             return
291     print("-" * 50)
292 
293 
294 # ========================================
295 # 6. CLI AND BULK PARSING LOGIC
296 # ========================================
297 
298 def parse_bulk_input(bulk_text):
299     """Safely extracts Token and Regex from raw text without the re
    module."""
300     # Normalize smart quotes that often occur from copy-pasting
301     normalized = bulk_text.replace('   ', '"').replace('   ', '"').
    replace("   ", "'").replace("   ", "'")
302 
303     extracted_specs = []
304     lines = normalized.strip().split('\n')
305 
306     for line in lines:
307         line = line.strip()
308         if not line:
309             continue
310 
311         # Split by quotes. A correctly formatted line like ("NAME", "
    REGEX")
312         # will yield at least 5 segments when split by '"'
313         parts = line.split('"')
314         if len(parts) >= 5:
315             name = parts[1]
316             regex = parts[3]
317             extracted_specs.append((name, regex))
318         else:
319             print(f"[!] Ignored malformed line: {line}")
320 
321     return extracted_specs
322 
323 
324 def main():
325     specs = []
326     dfa_ready = False
327     dfa_s, dfa_t, dfa_a = None, None, None
328 
329     while True:
330         print("\n=== Lexical Analyzer CLI ===")
331         print("1. Add Single Token Specification")
332         print("2. Add Bulk Token Specifications (Paste a list)")
333         print("3. Compile Lexer (Build DFA)")
334         print("4. Input Source Code & Run Lexer")
335         print("5. Exit")
336         choice = input("Select an option: ")
337 
338         if choice == '1':
```

```
            name = input("Enter Token Name (e.g., ID): ")
            regex = input("Enter Regex (e.g., [A-Za-z][A-Za-z0-9_]*): "
)
            specs.append((name, regex))
            dfa_ready = False
            print(f"Added {name} -> {regex}")

        elif choice == '2':
            print("Paste your bulk specifications below.")
            print("Format must be: (\"TOKEN_NAME\", \"regex\"),")
            print("Type 'DONE' on a new line and press Enter when
finished:")

            bulk_lines = []
            while True:
                try:
                    line = input()
                    if line.strip().upper() == 'DONE':
                        break
                    bulk_lines.append(line)
                except EOFError:
                    break

            new_specs = parse_bulk_input('\n'.join(bulk_lines))
            for name, regex in new_specs:
                specs.append((name, regex))
                print(f"Added {name} -> {regex}")
            if new_specs:
                dfa_ready = False

        elif choice == '3':
            if not specs:
                print("No specifications loaded!")
                continue
            super_start = State()
            for priority, (name, regex) in enumerate(specs, 1):
                try:
                    nfa = regex_to_nfa(regex)
                    nfa.accept.is_accepting = True
                    nfa.accept.token_type = name
                    nfa.accept.priority = priority
                    super_start.e_transitions.add(nfa.start)
                except Exception as e:
                    print(f"Error compiling {name} (Check regex syntax)
: {e}")
            dfa_s, dfa_t, dfa_a = build_dfa(super_start)
            dfa_ready = True
            print("DFA Compilation Successful.")

        elif choice == '4':
            if not dfa_ready:
                print("Please compile the Lexer (Option 3) first.")
                continue
            source = input("Enter source code string: ")
            simulate(source, dfa_s, dfa_t, dfa_a)

        elif choice == '5':
            print("Exiting...")
```

```
            break
        else:
            print("Invalid choice.")


if __name__ == "__main__":
    main()
```

Appendix B: User Manual

System Requirements

This application is written entirely in Python 3. It runs in terminal environments without requiring external libraries or frameworks (such as `re`), ensuring a pure theoretical implementation of automata-based parsing.

Getting Started

1. Save the source code (from Appendix A) into a file named `lexer.py`.
2. Open your command prompt or terminal.
3. Execute the script by typing: `python lexer.py`.

Menu Navigation

Upon starting the application, you will interact with the Main Menu.

Option 1 & 2: Loading Token Specifications

To build the Lexer, you must first define the rules. The system prioritizes rules sequentially—enter Keywords before Identifiers to ensure the Tie-Breaking mechanism resolves correctly.

- **Option 1:** Allows you to type a single Token Name and its Regex manually.
- **Option 2 (Recommended):** Allows you to paste a predefined list of tokens. The format strictly expects pairs separated by quotes, e.g., `("KW_IF", "if"),`. Paste the list, press Enter, type `DONE` on an empty line, and press Enter again to load them.

Regex Features Supported: Standard characters, grouping `()`, union `|`, closures (`*`, `+`, `?`), character classes (`[A-Za-z]`, `[0-9]`, `[A-Za-z0-9_]`), and escape sequences (`\+`, `\-`).

Option 3: Compile Lexer

Before scanning code, you must select Option 3. This command processes all loaded regex strings, converts them into NFAs, merges them into a Super NFA, and executes the Subset Construction algorithm to generate a deterministic transition table (DFA). You will receive a success message when the engine is ready.

Option 4: Scanning Code

Once compiled, select Option 4. Paste or type your raw source code and press Enter. The Lexer will apply Maximal Munch to process the string from left to right, ignoring whitespaces. It outputs a formatted table detailing the recognized Lexeme, the matching Token identity, and the exact Line and Column location. If an illegal character is encountered, a "Lexing Error" will halt the program and report the exact point of failure.